

To Every Action, An Opposite Reaction!

Planrs Academy Grade 3 Physics Master Manual • Duration: 30 Mins

The Story Hook: Sriharikota Countdown

Young Vihaan and his older sister Zarawar are sitting cross-legged on the rug with their grandfather (Dada-ji), watching a live television broadcast of a giant space launch!

The announcer booms: "Three... Two... One... Lift-off! The ISRO Chandrayaan vessel leaves the launch pad!"

A massive wave of bright orange fire and thick white smoke bursts out of the bottom nozzle, pushing down hard against the ground. At that exact second, the giant metal rocket climbs smoothly into the clouds!

Vihaan scratches his head and asks: "Dada-ji, the fire is pushing straight **DOWN**, so why is the heavy rocket going straight **UP** into the sky? How does pushing down help things go up?"

Newton's Double Push Rule

Dada-ji pauses the screen to share a secret rule discovered by a famous scientist named Sir Isaac Newton. In our world, forces never exist entirely on their own. They always work in matching pairs!

Newton's Third Law of Motion

"For every push in one direction, there is an equal push back in the opposite direction!"

Force Category	What it means	Physical Action
☒ The Action Force	The very first push you apply.	The element pushing backward or downward.
The Reaction Force	The instant matching response push.	The companion force pushing back in the opposite direction with the same strength.

Everyday Example: Stepping Off a Naav

Before building a rocket, let's look at a traditional wooden boat (Naav) on a beautiful Kerala canal:

← Action Force

Your foot pushes backward hard against the wooden boat hull to step off.

→ Reaction Force

The boat structure pushes your body forward onto the dry pier floor!

The Result: You glide safely onto the dock while the loose boat slides away into the water!

Concept Practice: Quick Check

Question 1: When you jump off the ground, your feet push down hard against the Earth (Action). What is the instant Reaction Force?

- A) The ground pushes your body upward into the air ✓
- B) The Earth moves backward out of the way
- C) The air pressure drops around you

Explanation: Your downward push on the ground creates an equal and opposite upward push from the ground onto you, sending you into the air!

Question 2: A bird flaps its wings downward against the air to stay aloft. In which direction does the air push back on the bird's wings?

- A) Downward
- B) Upward ✓
- C) Sideways

Explanation: Pushing the air down (Action) causes the air to instantly push the bird up (Reaction), keeping it flying.

Meet the Model: The Balloon Rocket

Let's build a mini spaceship using simple everyday classroom items:

- **Long String Path:** Stretched tight between two walls to keep our vehicle straight.
- **Plastic Straw:** Threaded on the string to let it slide smoothly with low friction.
- **Rubber Balloon:** Taped to the straw to act as our fuel engine container.

Lab Exploration Setup

Blow up a bright red balloon and hold it shut with a clip. When you release the clip, watch the forces split instantly:

← Action (Leftward)

The trapped air rushes out backward through the open nozzle at high speed.

→ Reaction (Rightward)

The balloon vehicle instantly zips forward across the string track!

Scaling Up to Real ISRO Spaceships

Now we can answer Vihaan's big question! A massive ISRO LVM3 Rocket works exactly like our classroom balloon:

↓ Spacecraft Action

Engines burn fuel to shoot a massive column of hot gas downward toward the pad.

↑ Spacecraft Reaction

An identical matching force pushes the heavy metal rocket upward into the sky!

The ISRO Launch Director Game

Imagine adjusting the nozzle variables inside a digital launch hangar simulation:

Test Run	Nozzle Direction	Flight Outcome
Hangar Test 1	Angled Sideways	The vehicle swerves off course and misses its target.
Hangar Test 2	Low Air Volume	A weak action push results in a very low, short flight.
Hangar Test 3	Straight Down	Max air push creates a perfect vertical flight into space!

Quick Concept Check

Question 3: If rocket engines push their exhaust gas toward the left side, which way will the vehicle fly?

- A) Backward (Leftward)
- B) Forward (Rightward) ✓**
- C) Straight Downward

Explanation: Because forces push back in opposite directions, a leftward push drives the rocket to the right!

Question 4: In our balloon rocket experiment, what specific element is the primary Action Force?

- A) Air rushing out backward ✓**
- B) The balloon zipping forward
- C) The long string guide track

Explanation: The air rushing out is the initial Action, which triggers the forward Reaction.

Your Home Physics Mission: The Skateboard Wall-Push!

Want to see this rule work around your own house? Try this simple playground test with an adult helper:

1. Stand balanced on a skateboard, roller skates, or a foot scooter near a sturdy brick wall.
2. Place both your palms flat against the surface of the wall.
3. Give the solid wall a firm, steady push forward away from your body.

Observe: Even though you push the wall forward (Action), your wheels instantly roll backward away from it (Reaction)! Sketch a quick diagram in your notebook using blue and gold arrows to show off your new physics skills!

Congratulations, Mission Directors! You've unlocked the core secrets of aerospace engineering!